RENNER ACADEMY 2025

How to assemble

and glue the RENNER hammerhead

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Course topics

- Unpack and install RENNER hammer shanks
- Mounting of RENNER hammershanks
- Alignement | compensate for distances | prepairs for traveling
- Traveling hammershanks
- Unpack Renner hammerhead set | count | sort | number
- Mark samples in the middle if there are no patterns
- Set sample hammerheads
- Enlarge the bore holes conically
- Principle of conical schank and conicale reamed hole
- Check hammer to shank angle
- Marking hammer position after samples
- Apply glue and gluing hammerheads
- Check gluing line and passage of the hammerheads
- Wich glue to use | hot vs. cold
- Control line of hammerheads | gluing of hammeheads
- Plastering the glue rings
- Clamping for plastering hammer tails
- Sawing protruding hammershanks
- Grind down | Sanding | Forming hammer tails
- Re-cleaning by hand | cosmetics

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Quick Overview

- Unpack the RENNER hammershanks
- Pre-install in fields according to the distribution of the instrument
- Let one or two pieces off between the fields
- Start with treble as packed by RENNER
- If used action rail, turn the screw at first in left direction to snap in place to the old thread pitch
- When found tighten the screw lightly with fingers
- Fix all hammershanks like this
- Our recommendation: use new RENNER screws for RENNER actions: for washers as disc springs, clean cross recess and combination slot and cross (maximum benefits)

Required tools:

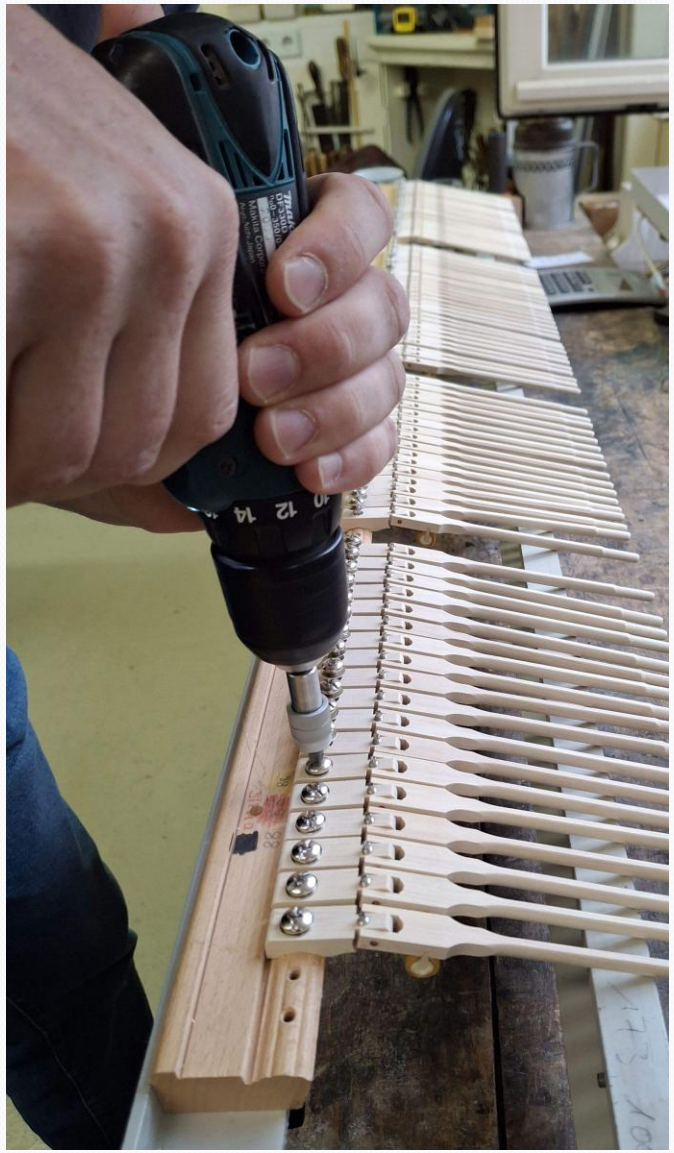
➢ GP action rack for RENNER agencys and repair workshops with different action rails for hammer gluing

Required material:

➢ Screw 4,5x22mm for flanges (grand shank use), with spring washer and combined screw slot Item number: 43-0008745

➢ RENNER hammerheads and hammershanks





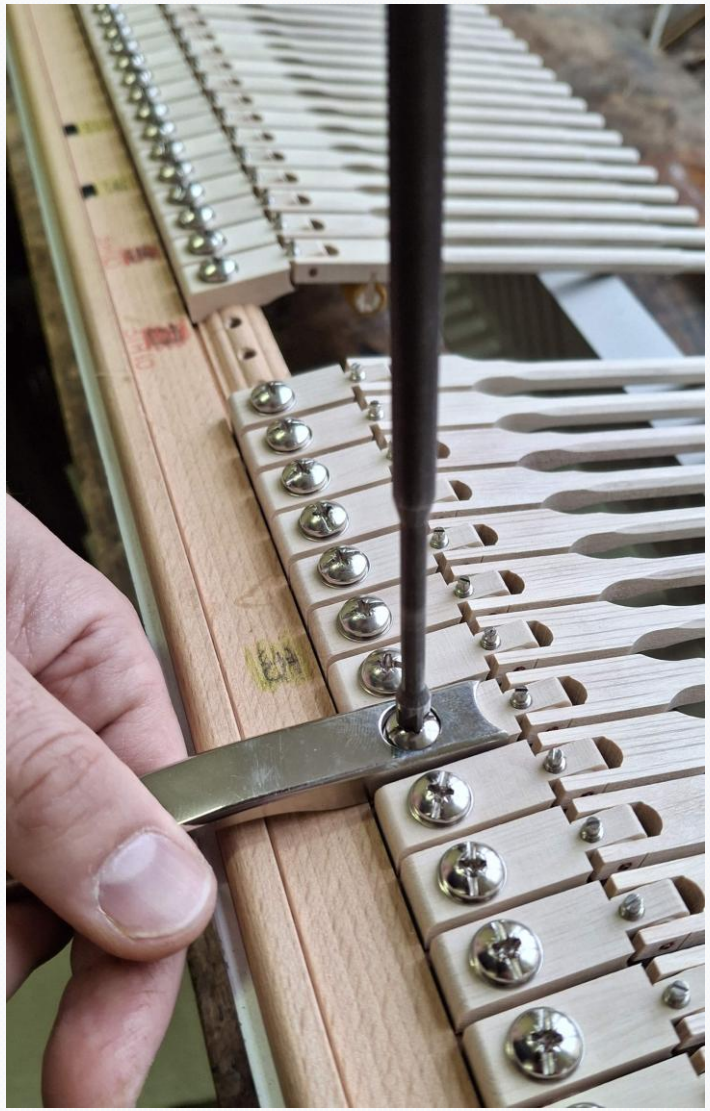
- Tighten screws
- ➢ Be carefull:
  do not tighten the screws to firmly
- This is the right moment to check the frequencies of the shanks
- lower sound for deeper notes
- higher frequencies for higher notes

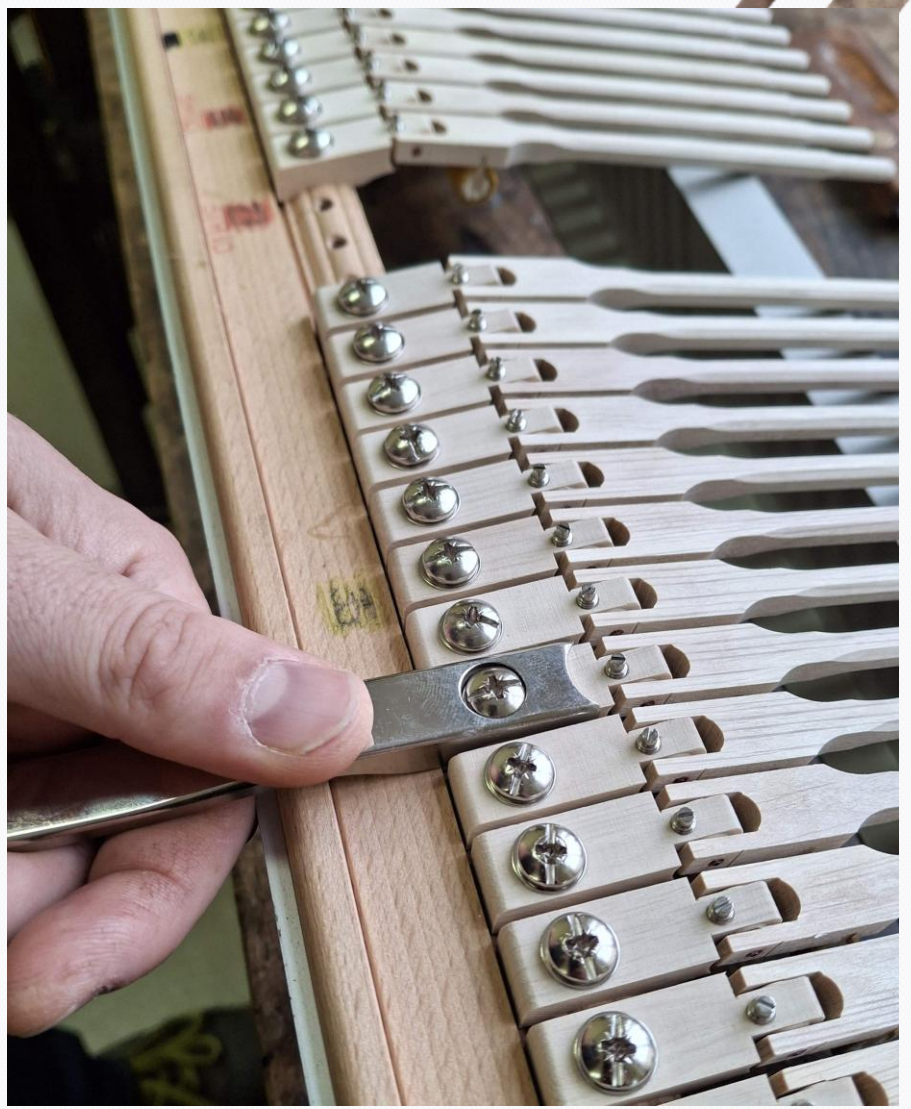
Required tools:

- ➢ battery screwdriver
- ➢ precision angle
- ➢ screw driver cross slot PZD 2 with ball handle RENNER Item number: 43-0008316
- ➢ Flange spacer Steinway and Baldwin, RENNER Item number: 43-0003286



- Put now the first and last hammershank of every field in 90° angle to the action rail





- Aligne the hammer shanks to each other
- Compensate different distances between them
- The better this work is done, traveling will be done

Required tools:

- Felt knife angled edge rough for, individual forming Cat.No.1739 RENNER item number: 43-0003252
- Travelling paper, one side adhesive, blue 0,2mm thick, 10 grams RENNER item number: 43-0005975



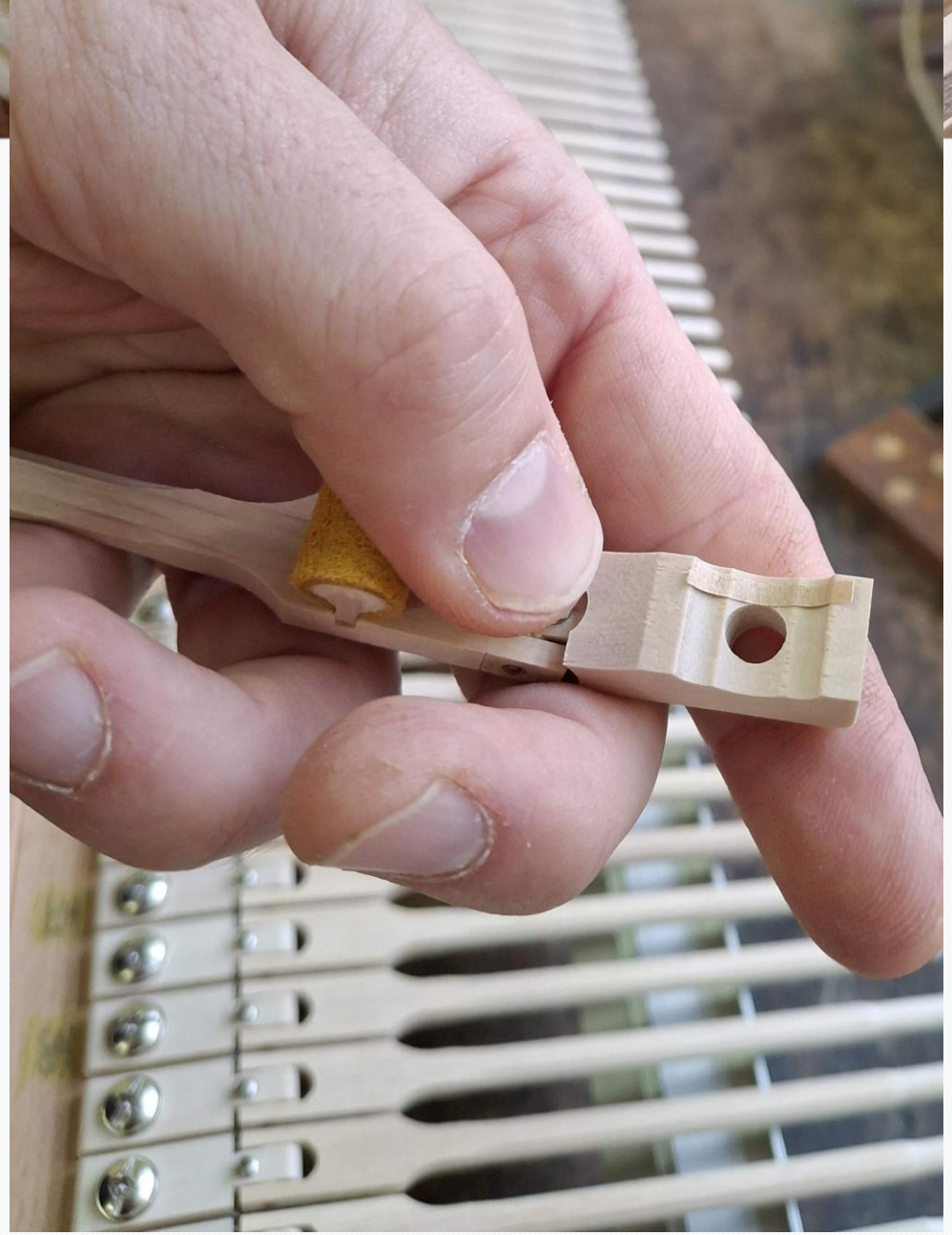


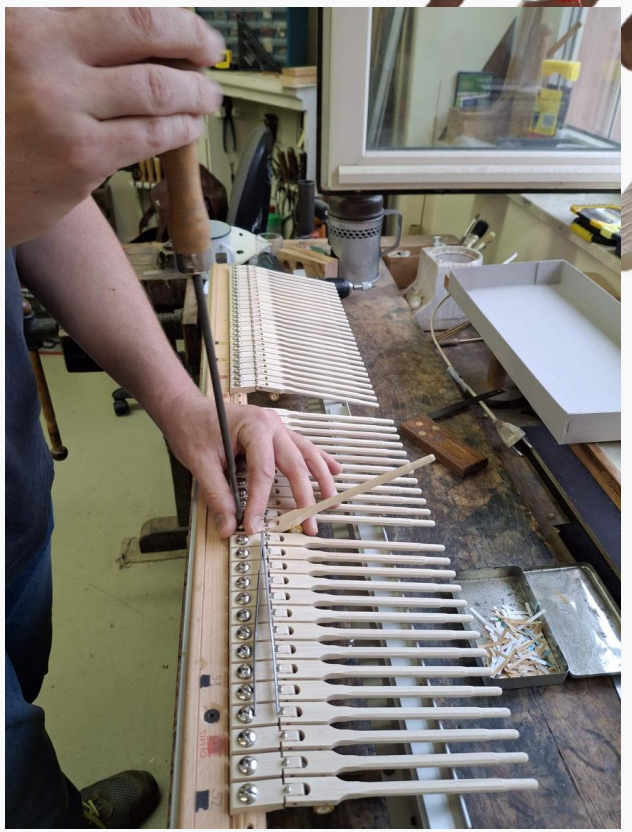




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Traveling hammershanks







➢ **Expert tip**

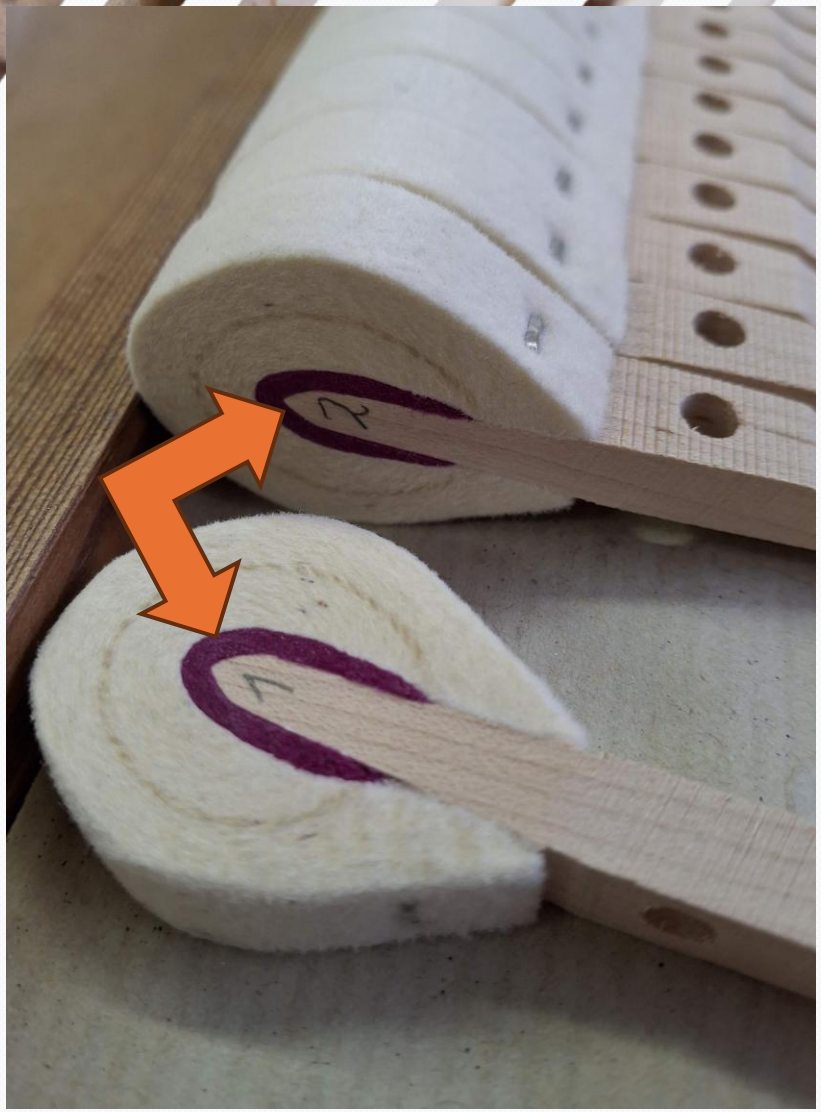
You can correct the way of the Hammers of a whole field in right or left direction if needed (this, you can find often at older grands and Steinway grands)



- Open the set pac
- Put the hammerheads on a working board
- Sort out the surplus hammerheads

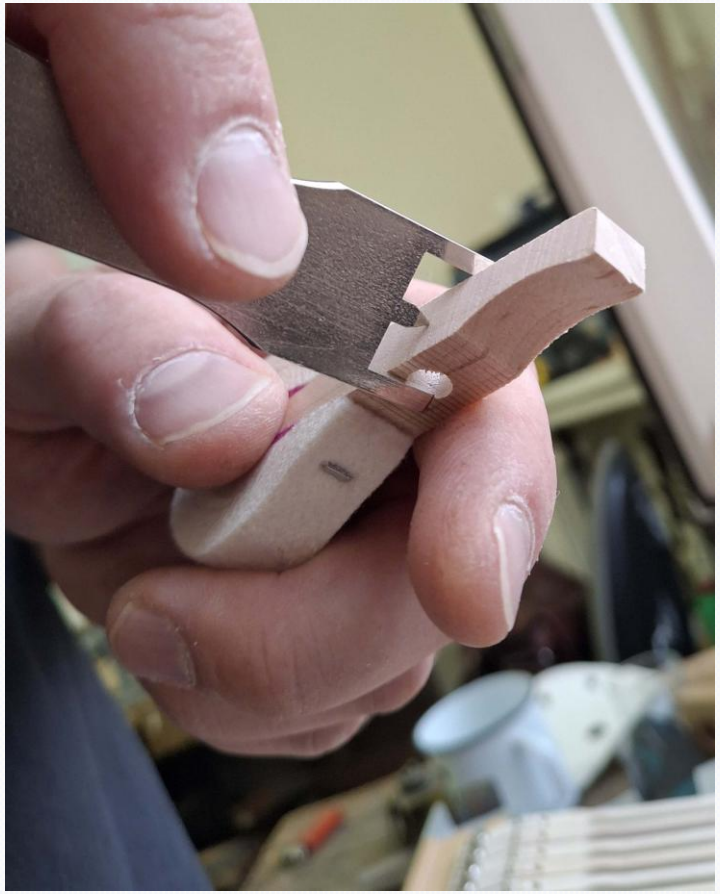
- Every RENNER hammerhead is signed with its number in the set and RENNER branding ex works (except original Steinway spareparts)
- If the hammerhead moulding is tapered (thinner then cutting width), sometimes the number is sanded away
- If you put out hammerheads when sorting before gluing numbers get lost

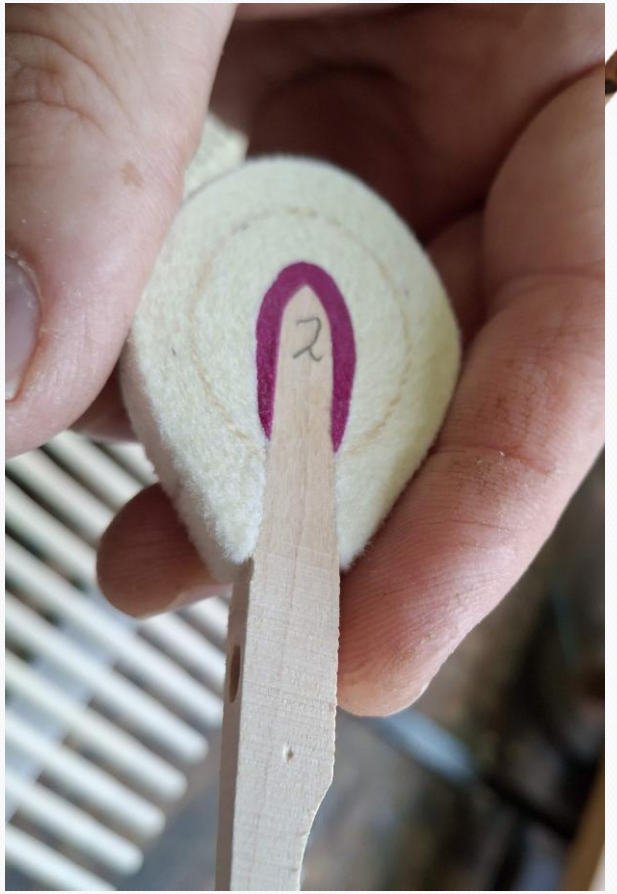
➢ **So notice:**
it is absolutly necessary to sign new numbers on the hammerheads

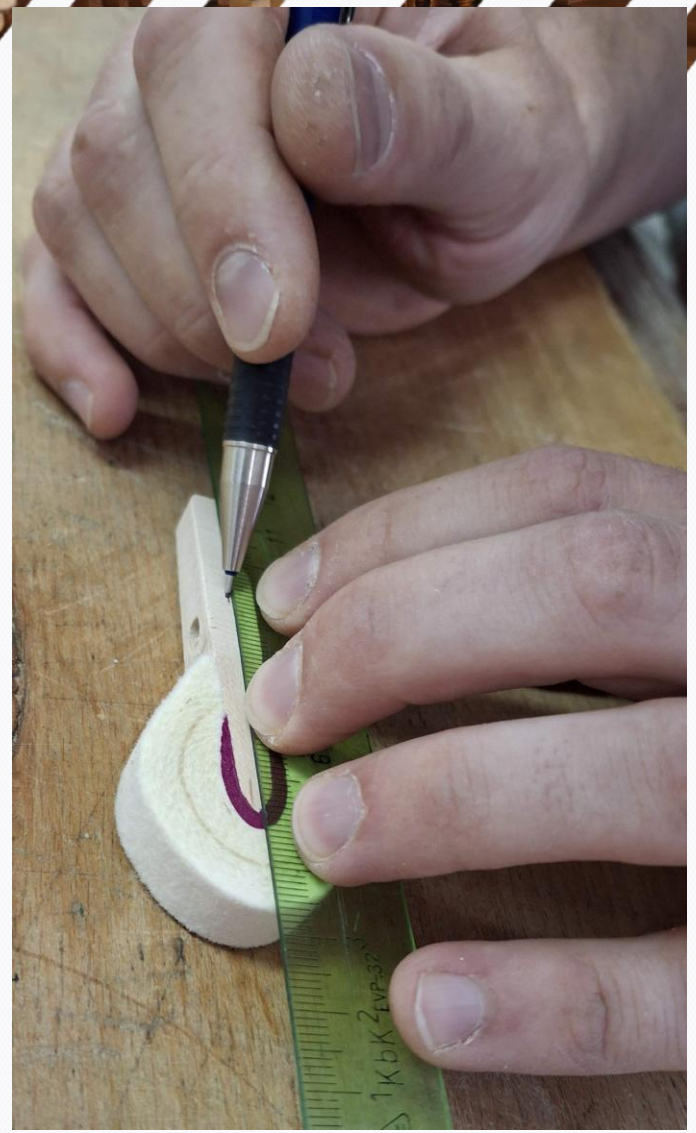


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Unpack Renner hammerhead set, count, sort and number





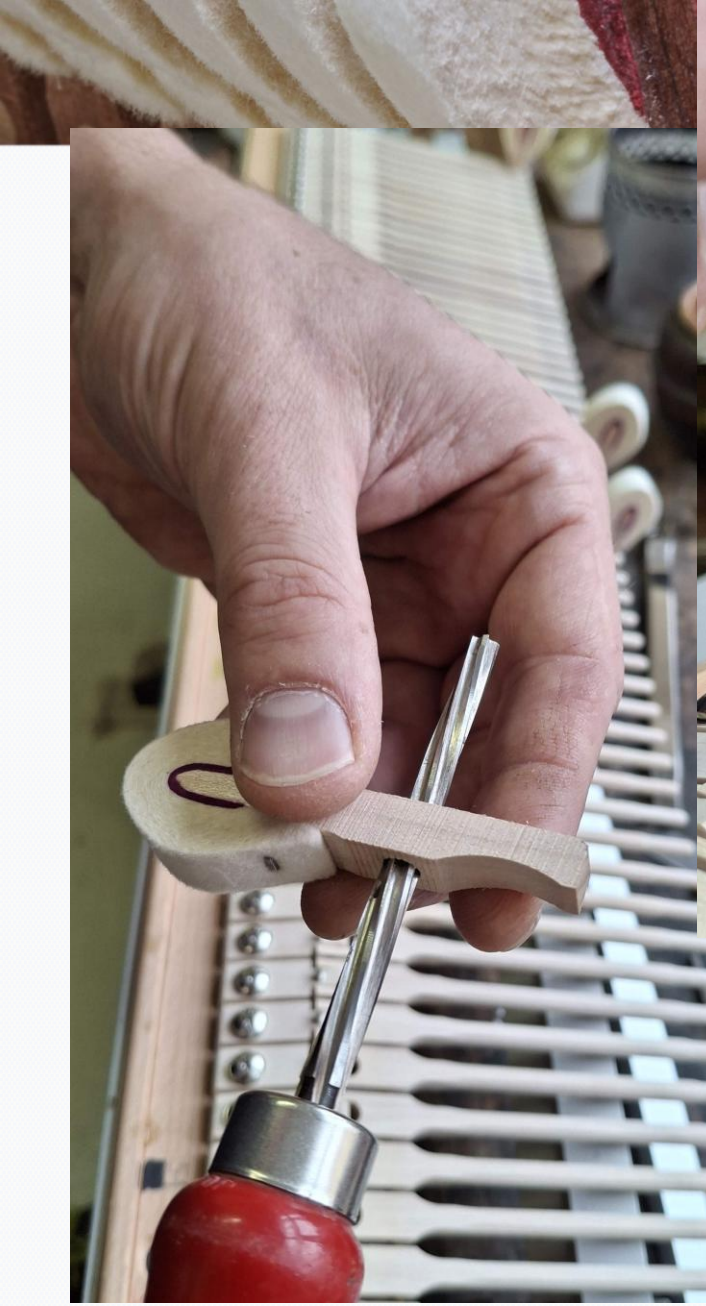


Required tools:

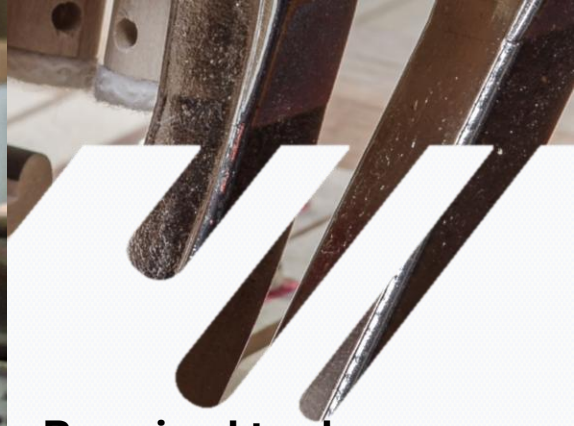
- Steel rule 150mm RENNER item number: 43-0003324
- Middle marking tool
- Mechanical pencil

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Mark samples in the middle if there are no patterns

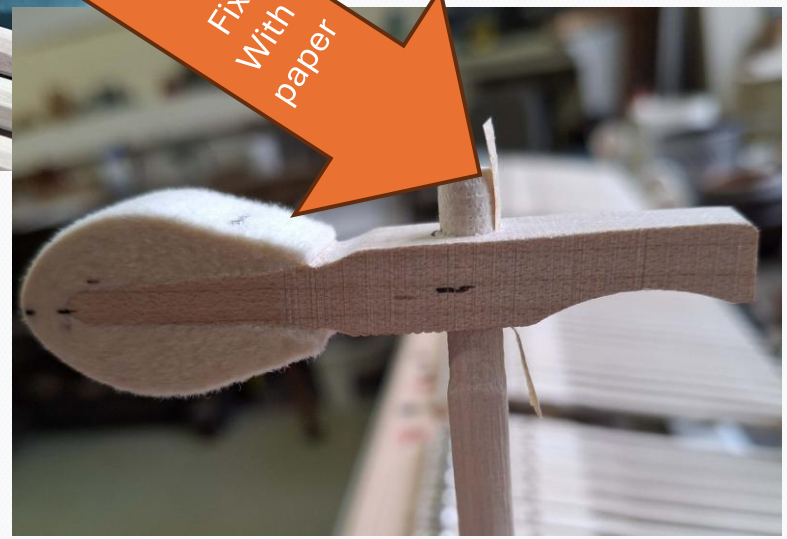






Required tools:

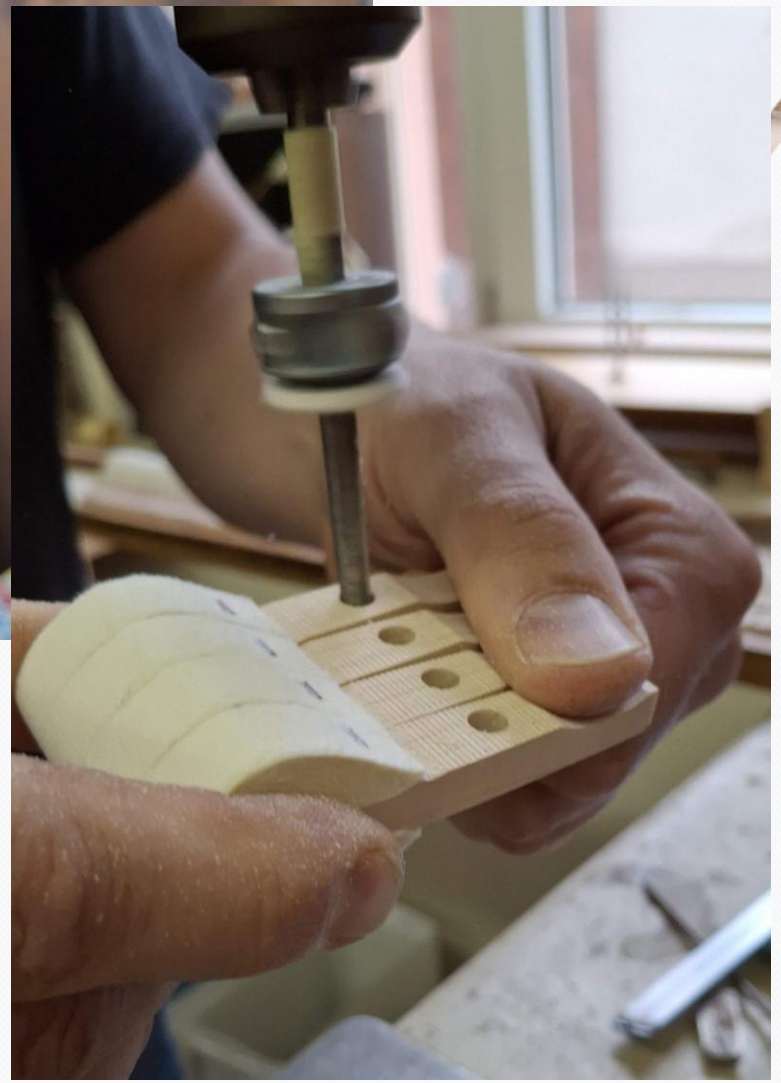
- Travelling paper, one side adhesive, blue 0,2mm thick, 10 grams RENNER item number: 43-0005975



Required tools:

- Hand reamer Ø5,0mm tapering size, 1:50 cutting direction clockwise RENNER item number: 43-0003409
- Circle





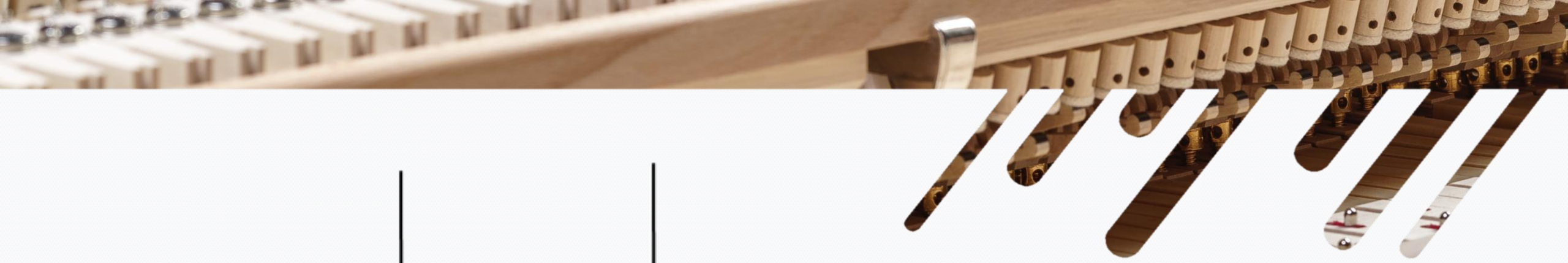


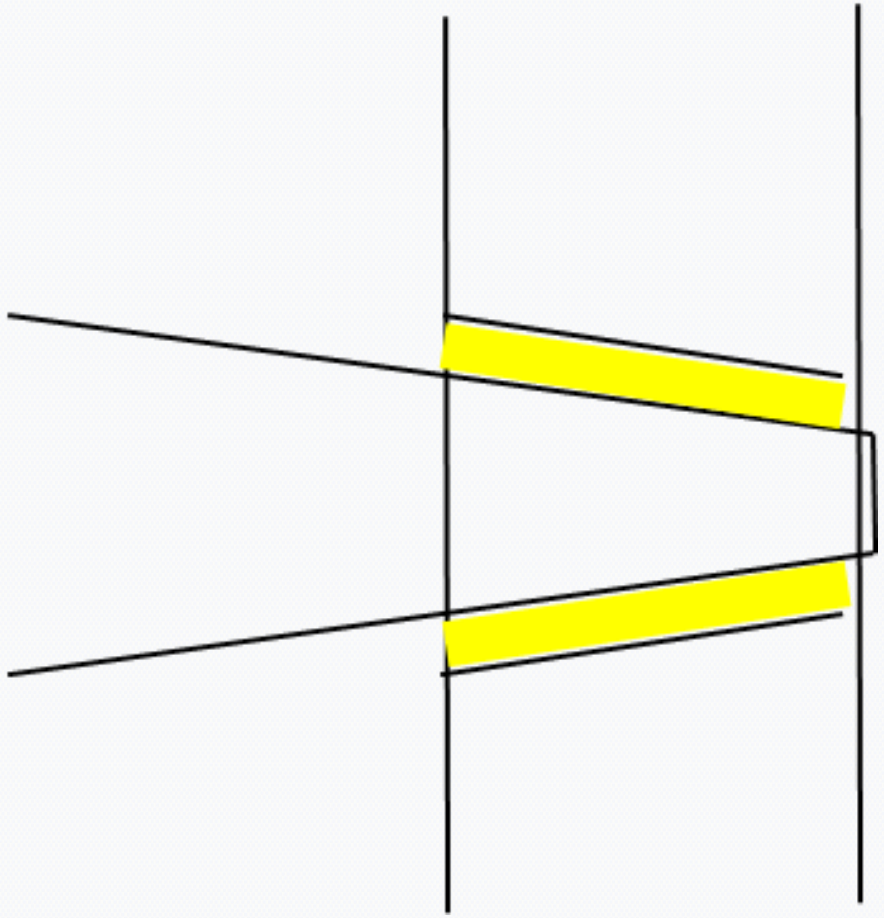
Required tools:

- Tapered reamer Ø5,0mm tapering size, 1:50 cutting direction clockwise RENNER item number: 43-0003409

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Enlarge the bore holes conically





- Conical hammer shank
- Conicle reamed bore hole
- Glue cannot be pushed trough
- The swelling of the wood quickly creates a firm connection

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Principle of conical schank and conicale reamed hole





- Set samples/ the first and last hammerhead at every field
- Check the hammer to shank angle

Required tools:

- Grand hammerhead rake gauge 91°, for measuring and correction during gluing process
- Item number: 43-0005650



Check hammer to shank angle





- Mark the hammer position makes gluing easier

Marking hammer position after samples



- Apply glue on the hammer shank



- Apply glue into boring
- Glue viscosity like liquid honey



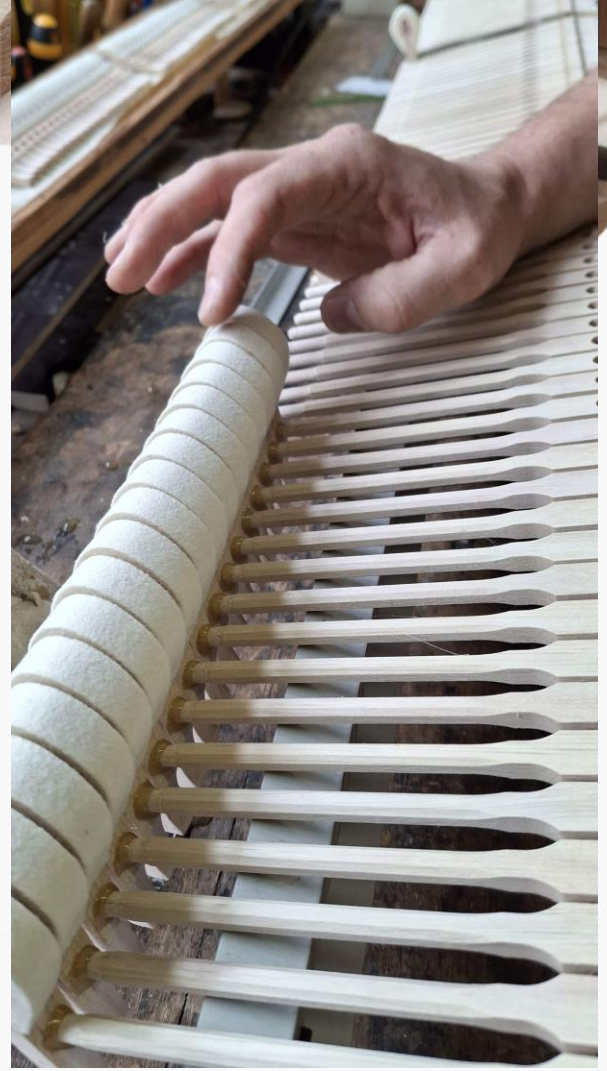
- Gluing hammer
- Rotary gluing for clean glue rings

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Apply glue and gluing hammerheads









- Glue 3 hammerheads and check the passage of the middle one
- Set the next hammer and check the passage of the hammer before
- Check the straight line of the hammers
- Check the straight line of the hammer tails
- Check the hammer to shank angle regulary

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Check gluing line and passage of the hammerheads







versus





63000 – 63080 Technische Daten für Warmleime im Vergleich

	Leim	Lieferform	Viskosität	pH	Bloomgrams
63000	Knochenleim [1]	Perlen	60-70 mPa•s [3]	5,75-7,0	160-180
63021	Körnerhautleim [1]	Gries	72-88 mPa•s [3]	5,5-7,5	235-265
63028	Hasengraupen [2]	Gries	5,25-5,75 [4]	5,5-6,5	340-360
63045	Pulvergelatine	Pulver	115-126 mPa•s [3]	5.8-7,0	340-380
63052	Hasenleim [2]	Platten	5,25-5,75 [4]	5,5-6,5	340-360
63060	Buchbinderleim [1]	Gries	100-120 mPa•s [3]	6-7	330-370
63080	Fischleim-Mischung	Gries	145-175 mPa•s [3]	6,0-7,0	575-625

[1] = aus Rindercollagen
[2] = Der Leim entspricht in seinen technischen Eigenschaften einem historischen Hasenleim.
[3] = mPa•s bei 12.50 % und 60°C
[4] = Viskosität in Engler

Gallertfestigkeit
Die Bloom-Zahl gibt die Elastizität eines Gallerte-Würfels an.

Viskosität
Die Millipascalsekunden geben die Auslaufzeit einer Leimlösung aus einem Trichter an.

Required tools:

- Electric glue pot, stainless steel, volume 0,2 liter, 220 volts
  Artikelnummer: 43-0003248

Required materials:

- Hide glue (pearl glue) brown
  Item number: 43-0001162
- Rabbit glue granules
  Item number: 43-0004644
- PVA glue, especially for hardwood
  Item number: 43-0007262



- Hammerhead felt line at the front- and back side



- Line of the hammer tails



- The entire hammer line with your eyes



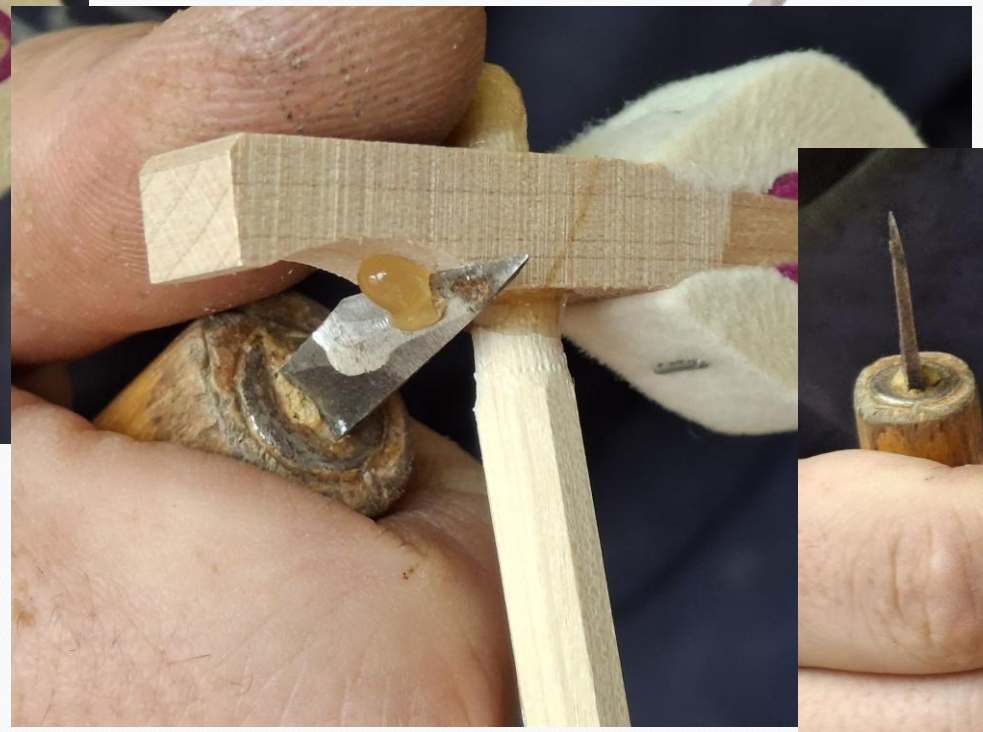


- If hot glue has been used, treat with a brush and hot water to ensure a perfect glue joint
- If cold glue has been used, you can do the same with water diluted cold glue (we call that „glue water“)

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Control gluing of hammeheads







- After approx one hour of drying time, the glue has the optimum strength (like vine gum) to plaster over any glue residue that has run

- Cut with a knife

- Put the rest away by hand

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Plastering the glue rings





Clamping for plastering hammer tails







Required tools:

- Cutting device clamp for grand hammershanks, assembled in the action after glueing the hammers
  Item number: 43-0003233

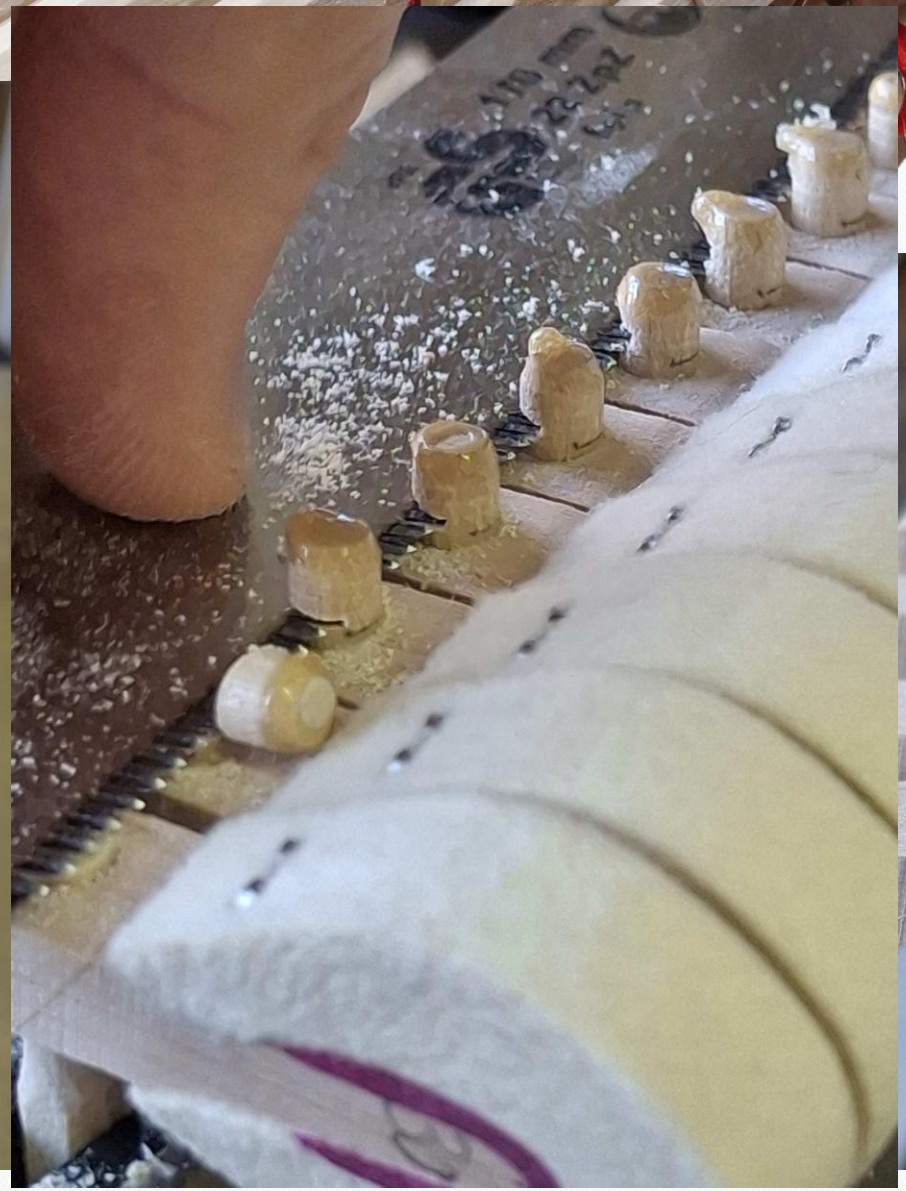


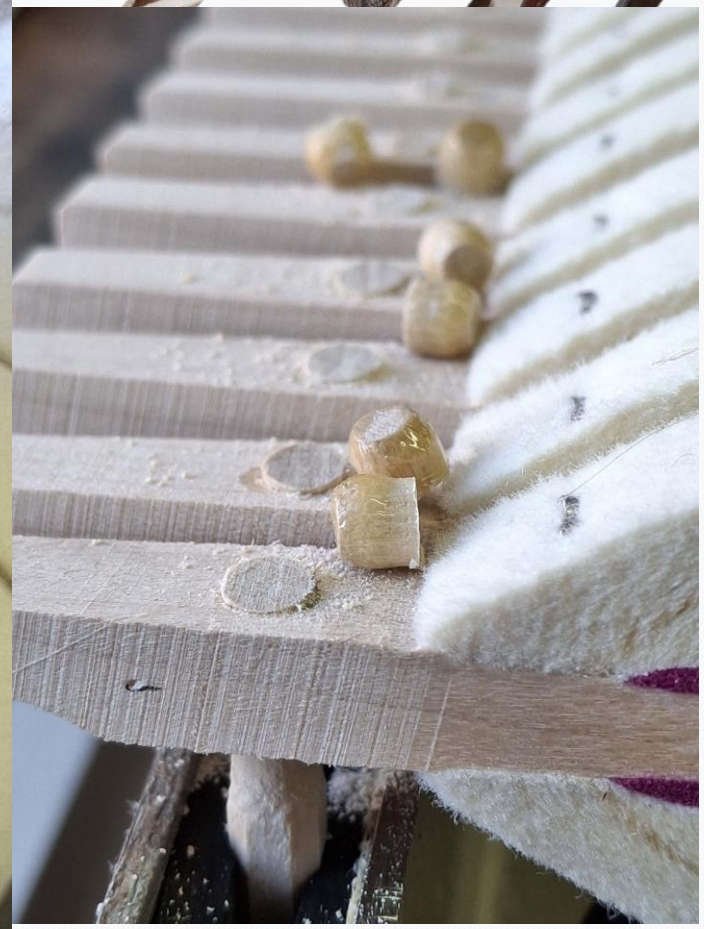
- Perfect clamping for plastering is necessary to protect the hammer shank axle

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Clamping for plastering hammer tails







- We use a Japanese saw for this

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Sawing protruding hammershanks





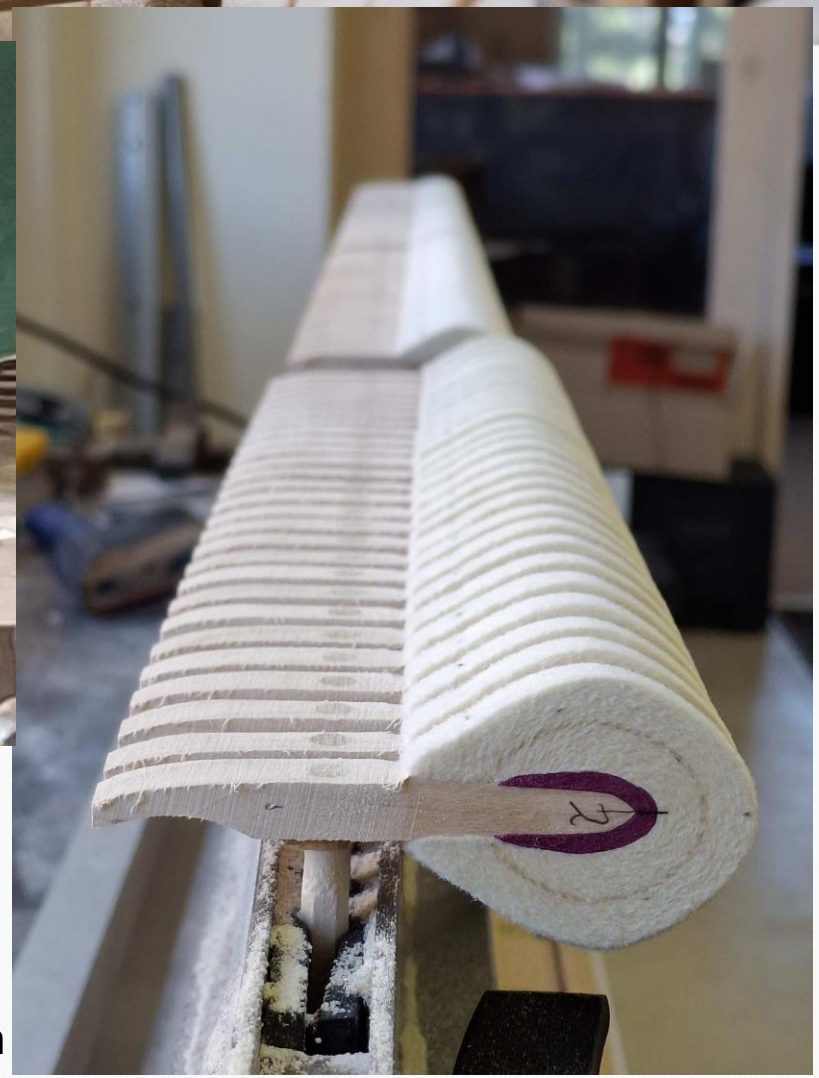
- Step 1: grind down overhangs



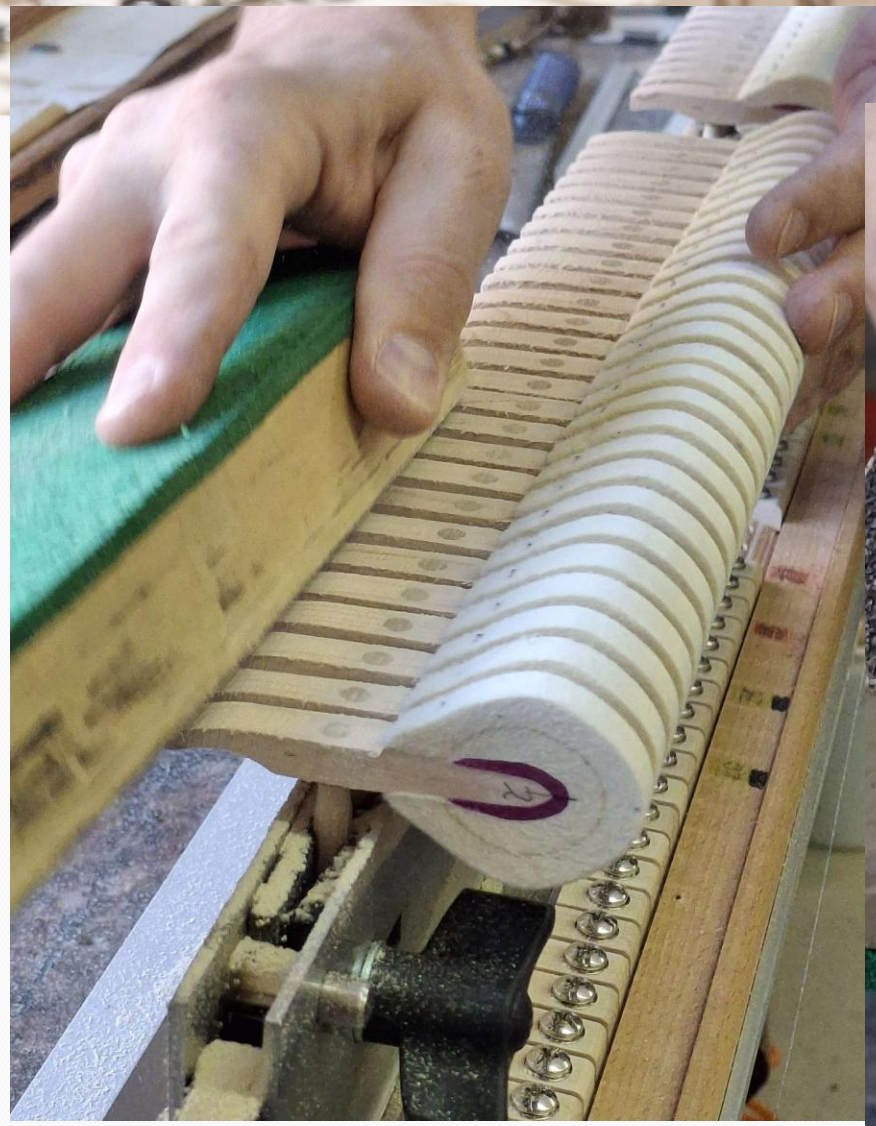
- Step 2: sanding an even surface



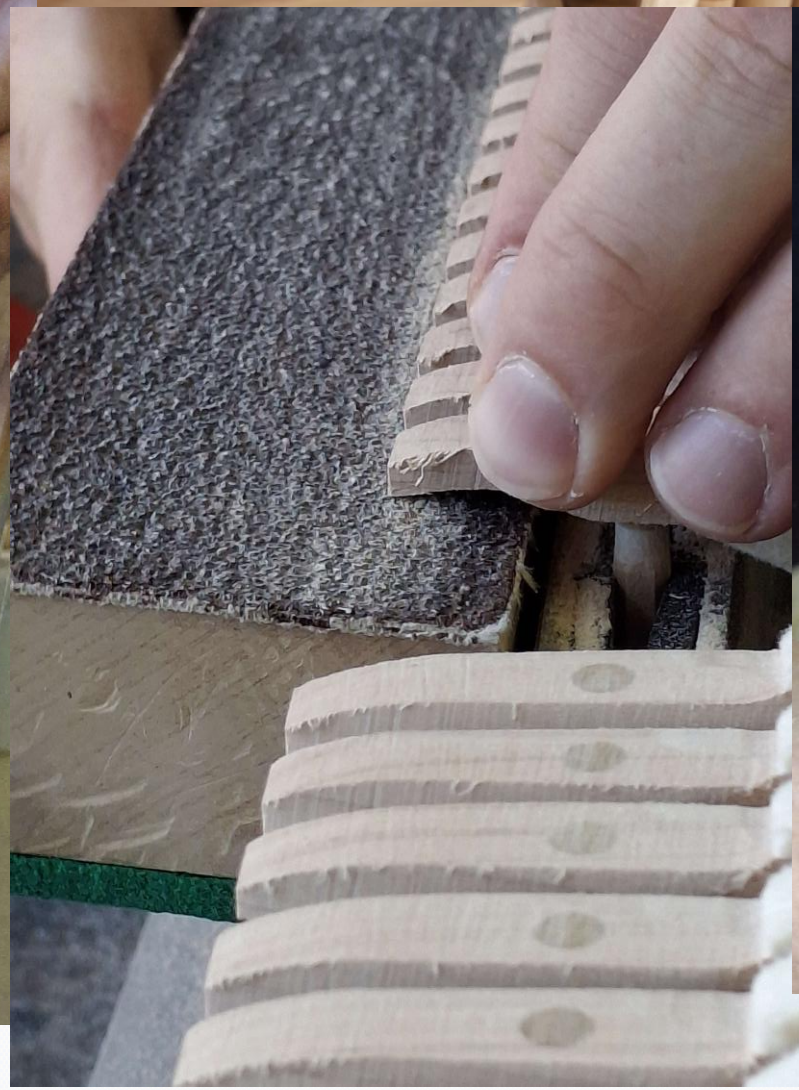
- Step 3: form out the hammer tail
- **Not to round** because can be pop up/ no good catch
- **Not to flat** because hooking possible

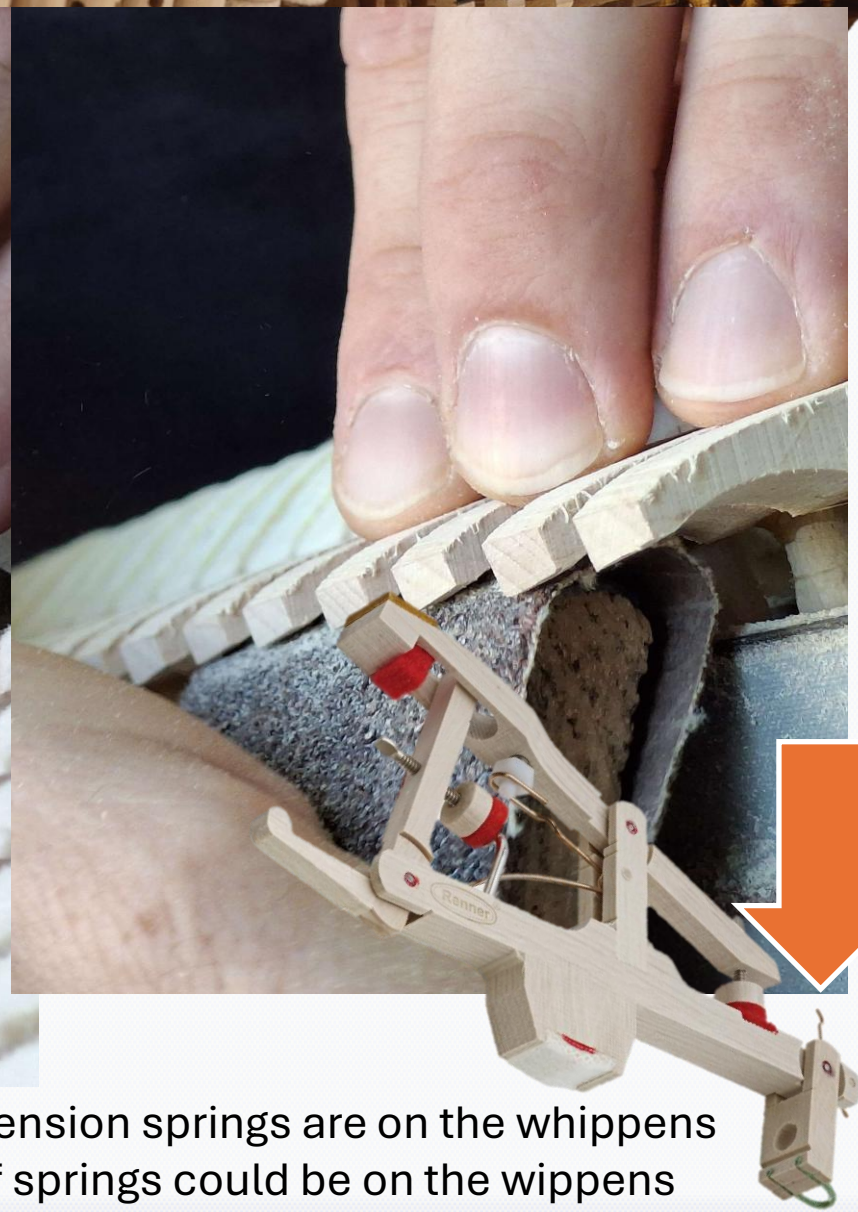


- Fully formed hammer tails



- Refinishing with a sanding block, grain size 40, to create a better grip



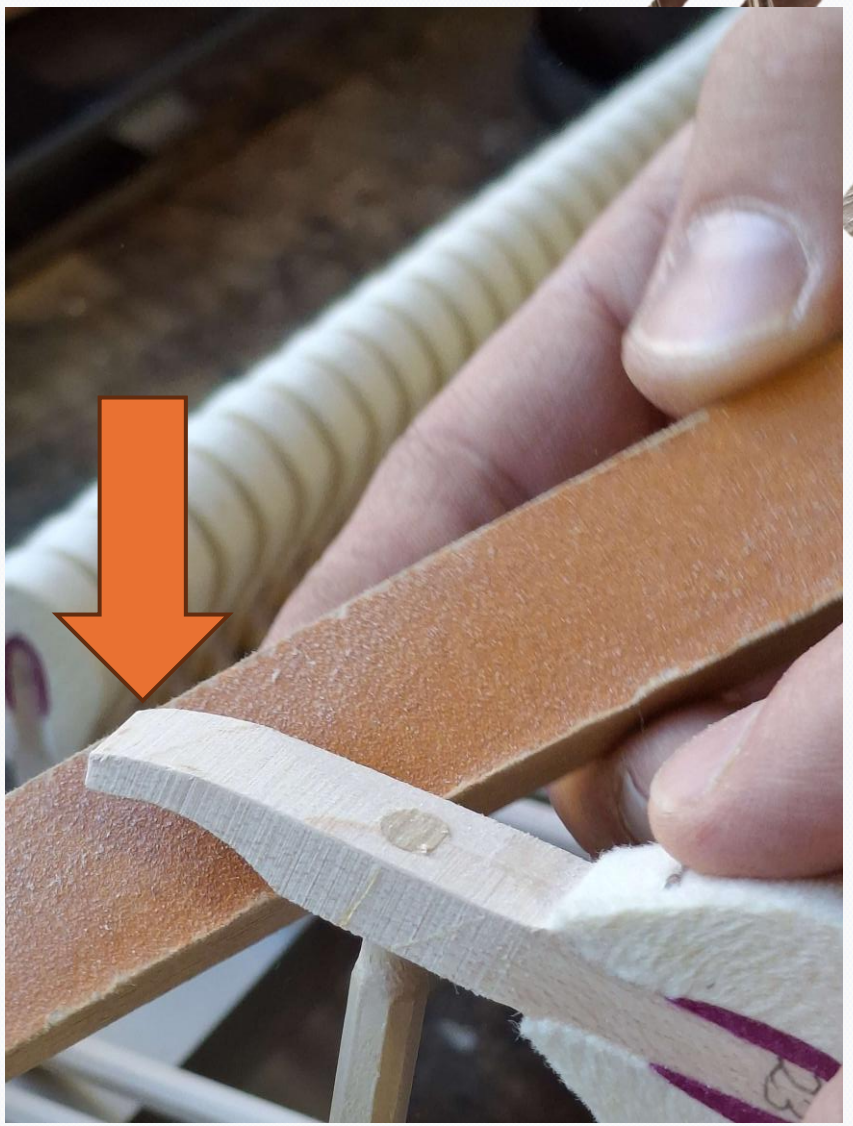


- Rework the back site edge if suspension springs are on the whippens
- Do this also, if you ´re not shure, if springs could be on the wippens

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Re-cleaning by hand





- Plaster with a fine sandpaper file, break the edges and corners
- Now work is done, and you have a last chance to correct passage of hammers with burner lamp
- Unscrew the hammershanks and pack for sending them to customer, or install them on the action and begin regulation works

Let the Perfect Hammer Sound!

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